Discrete Mathematics & Graph Theory Foundations

Course: Discrete Mathematics | Semester 4 | Contributor: Alex Chen

Propositional logic, recurrence relations, generating functions, and Euler Hamiltonian graphs.

Logic & Relations

* Equivalence relations, partial orders, Hasse diagrams, and Poset lattices.
* Pigeonhole principle and generalized pigeonhole applications in combinatorics.
* Solving linear recurrence relations with characteristic roots.

Graph Theory Highlights

* Handshaking lemma: sum of degrees of all vertices equals 2 * |E|.
* Planar graphs: Eulers formula V - E + F = 2 and Kuratowskis theorem K5 and K3,3.
* Euler circuits all even vertex degrees vs Hamiltonian cycles NP-complete.

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